Solution to Question 1 (30 marks)

a)

i) The inverter can generate voltage vectors that are within the hexagon formed by the voltages that the different possible switching states generate. This is a voltage in the range between using $\frac{\sqrt{3}}{3}V_{DC}$ (467 V peak) and $\frac{2}{3}V_{DC}$ (404 V peak) depending on the angle of the vector. To test if these voltages can be applied, we must calculate the norm of the voltage vectors in variables.

$\left\|v_1^{\alpha\beta}\right\| = 374$ V (which is within the range.)
$\left\|v_2^{\alpha\beta}\right\| = \left\|Cv_2^{abc}\right\| = 513$ V (which is out of the range)

[3 marks]

ii) A balanced three-phase voltage vector becomes a vector of constant modulus rotating at ω_e rate (ie at the angular frequency of the grid) in $\alpha\beta$ variables. $\frac{2}{3}V_{DC}$ is the radius of the largest circle inscribed in the hexagon; consequently, any voltage with a modulus smaller than this threshold won't hit the limit during a complete cycle of the voltage. In contrast, a voltage vector greater than $\frac{2}{3}V_{DC}$ would be trimmed to fit within the hexagon and consequently it would contain low-frequency distortion.

[2 marks]

iii) To answer this question we must find out in which region of the hexagon the vector belongs. We can obtain the angle of the vector as $atan\left(\frac{v_1^{\alpha}}{v_1^{\beta}}\right)$. This gives an angle of 15º, which indicates that the vector belongs to the first sector, ie within vectors ①, ②. Therefore, the converter will apply the switching states that correspond to vectors 0, 1, 2 and 7. 0 and 7 are the switching states that generate a zero voltage, which is required to make the modulus of the generated voltage be lower than that of switching states 1 and 2.

[2 marks]

iv) To answer this question we must solve the following equation:

$$v_1^{\alpha\beta} = \frac{1}{T_s}\left(t_{①}v_{①}^{\alpha\beta} + t_{②}v_{②}^{\alpha\beta}\right)$$

We can obtain the components of the vectors that correspond to the switching states from the diagram of the hexagon:

$v_{①}^{\alpha\beta} = \begin{bmatrix}\frac{2}{3}V_{DC}\\ 0\end{bmatrix} = \begin{bmatrix}466.7\\ 0\end{bmatrix}$ V and $v_{②}^{\alpha\beta} = \begin{bmatrix}\frac{1}{3}V_{DC}\\ \frac{\sqrt{3}}{3}V_{DC}\end{bmatrix} = \begin{bmatrix}233.3\\ 404.1\end{bmatrix}$ V

This corresponds to a linear system with two equations and it gives: $\frac{t_{①}}{T_s} = 0.65$ and $\frac{t_{②}}{T_s} = 0.24$. Therefore, the two vectors have to be applied by: 65 µs and 24 µs respectively. The remaining time in the cycle, ie 11 µs will be used to apply 0 voltage, this can be split evenly as 5.5 µs and 5.5 µs for the switching states of 0 and 7.

[5 marks]

b)

Model Answers

i) This is a fairly straightforward calculation: $I_{rated} = \frac{P_{rated}/3}{V_{AC}/\sqrt{3}} = 14.4$ A.

[3 marks]

ii) The voltage of the bridge can be calculated using the steady-state equation of the grid-side circuit, which is:

$$\bar{V}_{inverter} - V_{grid} = jX\bar{I}$$

We are given the X in pu, and it corresponds to X=4 Ω given that the Z_{base} is 14 Ω.
If I=15 A, then $\bar{V}_{inverter} = 230 + j\ 60$ V which gives V=239 V ph-n. Which corresponds to 413 V ph-ph rms.

[3 marks]

iii) The main difference between delivering reactive power or absorbing reactive power that can cause the ratings to be difference is in terms of the voltage that needs to be applied. This becomes apparent from the equation of the grid-side circuit:

$$\bar{V}_{inverter} = V_{grid} + jX\bar{I}$$

If the current is reactive, the rotation caused by j will make the voltage drop of the reactor to be aligned with the voltage of the grid. A capacitive reactive power setpoint will cause the voltage of the inverter to be greater than the voltage of the grid and more likely to hit the limit of the maximum voltage the inverter can generate.

[3 marks]

c)

i) A typical Grid-Following (GFL) controller uses the Park transformation to transform the instantaneous vectors of voltage and current from *abc* representation to *dq*. This makes the variables be constant in steady-state, which enables the use of PI controllers with high gain at zero frequency to control currents or voltages. It also requires a suitable angle for the transformation to be chosen, for GFL control this is obtained using a PLL fed with voltage measured at the point of connection.

[3 marks]

ii) The PLL has to find the appropriate angle for the Park transformation matrix that makes the q component of the voltage be zero. Typically, this is obtained using a closed loop that uses the q component as the error to be minimised; however, it is possible to calculate the angle as

$$\theta := atan\left(\frac{v_\beta}{v_\alpha}\right)$$

This would make the q component be strictly zero regardless of the presence of harmonics or phase imbalance.

[3 marks]

iii) Under steady-state balanced operation, the *dq* components of voltages and currents are constant values if the derivative of the angle of the Park transformation matches the angular frequency of the grid. However, if these contain negative sequence, this appears as sinusoidal terms of two times the angular frequency of the grid. This can cause the estimated angle and frequency of the grid to oscillate at this frequency if the controllers aren't designed properly.

[3 marks]

Model Answers

Solution to Question 2 (25 marks)

a)

i) The equilibrium can be calculated by setting $d\delta/dt = 0$ in the swing equation.

$$P'_{Mech} - \frac{EV_{IB}}{X_S}\sin\delta_0 = 0 \implies \delta_0 = \text{asin}\frac{P'_{Mech}X_S}{EV_{IB}} = \text{asin}0.2 = 0.201 \text{ rad} = 11.5 \text{ deg}$$

[2 marks]

ii) The linearised equation is get from Taylor expension around the equilibrium.

$$\frac{d^2\Delta\delta}{dt^2} = -\frac{K_D}{J}\frac{d\Delta\delta}{dt} - \frac{EV_{IB}\cos\delta_0}{JX_S}\Delta\delta$$

$$\frac{d^2\Delta\delta}{dt^2} + 2\frac{d\Delta\delta}{dt} + 9.80\Delta\delta = 0$$

[4 marks]

iii) The characteristic equation of the linearised swing equation is

$$Js^2 + K_D s + \frac{EV_{IB}\cos\delta_0}{X_S} = 0$$

Letting $J = 0$, we solve the undamped natural frequency

$$\omega_n = \sqrt{\frac{EV_{IB}\cos\delta_0}{JX_S}} = \sqrt{\frac{\cos 0.201}{0.5 \times 0.2}} = 3.13 \text{ rad/s}$$

The natural frequency is negatively correlated to J.

[4 marks]

iv) Solving the characteristic equation yields the pole:

$$s = \frac{-K_D}{2J} \pm j\sqrt{\frac{EV_{IB}\cos\delta_0}{JX_S} - \frac{K_D^2}{4J^2}} = \frac{-1}{2 \times 0.5} \pm j\sqrt{\frac{\cos 0.201}{0.5 \times 0.2} - \frac{1^2}{4 \times 0.5^2}} = -1 \pm j2.967$$

It is clear that the damping helps to move the pole towards the left halp plane and therefore stablise the swing.

[4 marks]

v) The damping in grid-forming inverters does not create losses, since the damping is emulated by control. The damping power comes in a way that the effective power from the resource (renewable or storage) is less than the setting value.

[2 marks]

Model Answers

b)

i) The resonant frequency is:

$$f_{res} = \frac{1}{2\pi\sqrt{\frac{L_1 L_2}{L_1 + L_2}C}} = \frac{10^6}{2 \times 3.14 \times \sqrt{\frac{200 \times 50}{250} \times 160}} = 1.99 \text{ kHz}$$

The transfer function of time delay is $e^{-j2\pi f T_d}$, and the phase shift is $2\pi f T_d$.

Given 1.5 cycle time delay, $T_d = 1.5T_s = \frac{1.5}{10 \times 10^3} = 0.15\text{ms}$.

At the resonant frequency, the phase shift is $2\pi f_{res} T_d = 2\pi \times 1.99 \times 0.15 = 107.46°$

The phase shift is higher than 90 degs so the inverter's current loop is not stable (the real part of the equivalent resistance becomes negative).

[4 marks]

ii) To make the inverter stable, the resonant frequency must be smaller than 1/6 of swiching frequency. The is

$$f_{res} = \frac{1}{2\pi\sqrt{\frac{L_1 L_2}{L_1 + L_2}C}} < \frac{1}{6} f_{sw}$$

From this we solve

$$C >= \frac{L_1 + L_2}{L_1 L_2}\left(\frac{3}{\pi f_{sw}}\right)^2 = \frac{250 \times 10^6}{200 \times 50} \times \left(\frac{3}{3.14 \times 10000}\right)^2 = 228 \text{ uF}$$

[2 marks]

iii)

- Reduce time delay. Update the PWM modulator immediately after sampling, without extra wait cycle.
- Reduce the resonant frequency. Ensure f_res<1/(4T_d) to avoid the negative damping.
- Reduce control gain. This will reduce the negative damping presented by the controller, but at the expense of reducing current regulation speed.
- Phase-leading compensation. Use derivation and prediction to compensate for the phase shift in time delay.
- Passive damping. Use extra resistor in the circuits to damp the resonance. The most common one is resistor in series with capacitor.
- Active damping. Use control to mimic the effect of passive damping.

[3 marks]

Model Answers

Solution to Question 3 (15 marks)

a) The bus incidence matrix is

$$T_{bl} = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & -1 \\ -1 & 0 & 0 & 0 \end{bmatrix}$$

[2 marks]

$$T_{BL} \times I_L = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & -1 \\ -1 & 0 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} I_{L1} \\ I_{L2} \\ I_{L3} \\ I_{L4} \end{bmatrix} = \begin{bmatrix} I_{L1} + I_{L2} + I_{L4} \\ -I_{L2} + I_{L3} \\ -I_{L3} - I_{L4} \\ -I_{L4} \end{bmatrix} = \begin{bmatrix} I_{B1} \\ I_{B2} \\ I_{B3} \\ I_{B4} \end{bmatrix} = I_B$$

[1.5 marks]

$$T_{BL}{}' \times V_B = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 1 & 0 & -1 & 0 \end{bmatrix} \times \begin{bmatrix} V_{B1} \\ V_{B2} \\ V_{B3} \\ V_{B4} \end{bmatrix} = \begin{bmatrix} V_{B1} - V_{B4} \\ V_{B1} - V_{B2} \\ V_{B2} - V_{B3} \\ V_{B1} - V_{B3} \end{bmatrix} = V_{\text{diff}}$$

[1.5 marks]

b)

i)

- The impedance seen by the IBR is relatively high when one transmission line is out of service.
- The IBR in grid-following mode controls the injected current $I\angle\varphi$ based on terminal voltage $V\angle\theta$ for power control while $V\angle\theta$ in turn depends on $I\angle\varphi$ and the grid equivalent impedance and voltage as

$$V\angle\theta = V_s\angle 0 + Z_{eq} \times I\angle\varphi$$

- For high effective impedance, $V\angle\theta$ is more sensitive to $I\angle\varphi$ which could destabilize the system

[1 mark each]

ii)

- The impedance seen by the IBR is relatively low when both transmission lines are in service.
- The IBR in grid-forming mode controls the terminal voltage $V\angle\theta$ based on injected current $I\angle\varphi$ for power (droop) control while $I\angle\varphi$ in turn depends on $V\angle\theta$ and the grid equivalent impedance and voltage as

$$I\angle\varphi = \frac{V\angle\theta - V_s\angle 0}{Z_{eq}}$$

- For low effective impedance, $I\angle\varphi$ is more sensitive to $V\angle\theta$ which could destabilize the system

[1 mark each]

Model Answers

iii) The instabilities can be dealt with by slowing down the IBR control loops.

[1 mark]

c)

- The q-axis voltage control loop is meant to force the q-axis component of the terminal voltage V_q to zero.
- A grid-forming converter sets the phase angle θ of its terminal voltage i.e. d-axis is automatically aligned with the terminal voltage
- The q-axis voltage control loop in a way mimics the action of a PLL which is not needed for a GFM making the q-axis loop redundant.

[1 mark each]

Model Answers